The discrete critical coagulation–fragmentation conjecture

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Part (ii), the case m = 1, part (i) as a corollary, final manuscript: Claude Opus 5.5 (effort xhigh)
Novelty audit, stationary singularity analysis, first draft: GPT-6-Astra via Codex CLI (effort xhigh)
Independent verification: Claude Opus 5.5 (effort xhigh)

Abstract. Jang and Tran conjectured that the discrete coagulation–fragmentation equation with multiplicative coagulation kernel ij and constant fragmentation kernel has, in the critical mass window $1/2 \le m < 1$, a unique stationary state of mass m given by an explicit recursion, and a unique global mass-conserving classical solution. We prove both statements, for every mass $0 < m < 1$, and the time-dependent one also for $m = 1$; the initial second moment assumption is not needed, only finite mass. With Jang and Tran's theorem that no global mass-conserving solution exists for $m > 1$, this settles the discrete critical-mass conjecture, gelation for $m > 1$ being understood in that sense. The zeroth-order coefficient of the associated Hamilton–Jacobi equation is $1/\tau(x)$ with $\tau(x) = 2\tanh(x/2)$. With x replaced by τ, Van's continuous half-slope invariant has a discrete analogue that propagates along characteristics with a nonnegative source. It yields a curvature barrier for all $m \le 1$, hence a global classical solution of the Hamilton–Jacobi equation by characteristics, with slope m at the boundary. Comparison with positive finite truncations then excludes mass loss; no viscosity-solution theory is used. Once mass is conserved the system is lower triangular, which gives uniqueness, convergence to the stationary state, and its positivity and mass identity. The stationary coefficients decay like $c\,R^{-n}n^{-5/2}$ with $R > 1$. The proofs have been checked by an independent verifier; they have not been refereed.

The question

The source is Jang and Tran [1]. The system is, for cluster densities $\rho_n = \rho(n,t) \ge 0$, $n \in \mathbb{N}$,

$$\dot\rho_n = \frac{1}{2}\sum_{i=1}^{n-1} i(n-i)\rho_i\rho_{n-i} - n\rho_n\sum_{j\ge1} j\rho_j - \frac{n-1}{2}\,\rho_n + \sum_{j>n}\rho_j, \tag{1}$$

which is their (1.1) with kernels (1.2), $a(i,j) = ij$, $b(i,j) = 1$. Their assumption (1.3) is $\sum_n n^2\rho_0(n) < \infty$, and their recursion (1.9) is

$$\tilde\rho_n = \frac{2m(1-m) + \sum_{i=1}^{n-1} i(n-i)\tilde\rho_i\tilde\rho_{n-i} - 2\sum_{i=1}^{n-1}\tilde\rho_i}{(2m+1)n+1}, \qquad n \ge 1. \tag{2}$$

Conjecture 2 of arXiv:2409.17974v1 (printed p. 7, end of Section 1.4), with notation typeset equivalently:

> *Conjecture 2. Consider (1.1) with kernels (1.2) and initial data satisfying (1.3). (i) For $1/2 < m < 1$, (1.1) admits a unique stationary solution $\tilde\rho : \mathbb{N} \to [0,\infty)$ such that $m_1(\tilde\rho) = m$. Further, $\tilde\rho$ satisfies the recursive formula (1.9). (ii) For $1/2 \le m_1(\rho_0) < 1$, (1.1) admits a unique mass-conserving solution ρ for all time. This solution ρ is in fact a classical solution.*

Jang and Tran proved existence for $m < 1/2$ under (1.3) (their Theorem 1.3), stationary states for $m \le 1/2$ (Lemma 5.7), uniqueness of mass-conserving solutions for $m \le 1$ (Proposition 1.2) and non-existence of global mass-conserving solutions for $m > 1$ (Theorem 1.1). Their Conjecture 1, which they attribute to [2] [3], asserts, for initial data satisfying (1.3), that $m = 1$ is the critical mass: gelation for $m > 1$, a unique global mass-conserving solution for $0 < m \le 1$. For continuous sizes, Tran and Van [4] proved existence for $m < 1/2$, and Van's preprint [5] extends existence to the window $1/2 \le m \le 1$; Van writes, in Section 1.2, p. 4: "so the discrete analogue of the window $1/2 \le m \le 1$ remains open as of this writing." The novelty audit is recorded in `references.md`; no earlier solution was found.

Notation and solution concepts. $m_k(\rho) = \sum_n n^k \rho_n$; $K(\rho)_n$ is the right side of (1), with the actual first moment in the coagulation loss; $\ell^1_1 = \{u : \sum n|u_n| < \infty\}$. For bounded $g : \mathbb{N} \to \mathbb{R}$, Jang and Tran's weak form (1.5) reads

$$\frac{d}{dt}\sum_n g(n)\rho_n = \frac{1}{2}\sum_{j,k}(g(j+k)-g(j)-g(k))jk\rho_j\rho_k - \frac{1}{2}\sum_j\sum_{k=1}^{j-1}(g(j)-g(k)-g(j-k))\rho_j. \tag{3}$$

Their Definition 1 calls ρ a solution if (3) holds for every bounded g, without fixing the time regularity. We call ρ a *mass-conserving solution* with datum ρ^0 if $\rho(t) \ge 0$, $m_1(\rho(t)) = m_1(\rho^0)$ for all $t \ge 0$, and for every bounded g the function $t \mapsto \sum g(n)\rho_n(t)$ is absolutely continuous on compact intervals and satisfies (3) for almost every t. This is the weakest reading; uniqueness is proved in this class. It is *classical* if every ρ_n is C^1 on $[0,\infty)$ and (1) holds for every $t \ge 0$. A *stationary solution* is a time-independent mass-conserving solution, that is, $\tilde\rho \ge 0$, $m_1(\tilde\rho) < \infty$, $K(\tilde\rho) = 0$.

Theorem 4 (global well-posedness). Let $\rho^0 \ge 0$ with $m = \sum_n n\rho^0_n \in (0,1]$; no other moment is assumed. Then (1) has exactly one mass-conserving solution with datum ρ^0. It is classical, every component is C^∞ on $[0,\infty)$, $\rho \in C([0,\infty);\ell^1_1)\cap C^1([0,\infty);\ell^1)$, the weak form (3) holds with a classical derivative at every $t \ge 0$ for every bounded g, and

$$\sum_n \rho_n(t) = m(1-m) + e^{-t/2}\Big(\sum_n \rho^0_n - m(1-m)\Big).$$

Theorem 5 (stationary states and relaxation). Let $0 < m < 1$. The numbers (2) are strictly positive, with $\sum_n \tilde\rho_n = m(1-m)$ and $\sum_n n\tilde\rho_n = m$; $\tilde\rho$ is a stationary solution and the only stationary solution with $m_1 = m$. Every mass-conserving solution of mass m converges to $\tilde\rho$ in ℓ^1_1. For $m = 1$, every component of the solution of Theorem 4 tends to 0 while its mass stays equal to 1.

Theorem 6 (decay of the stationary state). For $0 < m < 1$ the generating function $\sum_n \tilde\rho_n z^n$ has radius $R = R(m) \in (1,\infty)$, and

$$\tilde\rho_n = \frac{b}{2\sqrt{\pi}}\, R^{-n} n^{-5/2}(1+O(n^{-1})), \qquad b^2 = \frac{(R-1)^2(m+\frac{1}{2})+\frac{3}{2}R}{R^2-1}.$$

Thus Conjecture 2 holds: part (i) is Theorem 5 and part (ii) is Theorem 4, both for every $0 < m < 1$, and part (ii) without (1.3). Theorem 4 at $m = 1$, together with Jang and Tran's Theorem 1.1 for $m > 1$, settles their Conjecture 1 in the discrete setting. Here "gelation for $m > 1$" is understood as in their Theorem 1.1: no solution conserves mass for all time. This also follows from Proposition 7, whose derivation of $n_0(t)$ holds on any interval $[0, T]$ and makes n_0 vanish at $T = 2\log(1 + n_0(0)/(m(m-1)))$, so no solution conserves mass on such an interval. Whether global weak solutions that lose mass exist for $m > 1$ is not addressed here. The status is *solved*: the proofs were checked by an independent verifier (file `verification.md`); they have not been refereed.

Strategy. Write $F(x,t) = \sum_n (1 - e^{-nx})\rho_n(t)$ (Jang and Tran's discrete Bernstein transform). The zeroth-order coefficient of its Hamilton–Jacobi equation (Jang–Tran (1.7)) is $\frac{1}{2} + \frac{1}{e^x - 1} = \frac{1}{2}\coth\frac{x}{2} = 1/\tau(x)$, $\tau(x) = 2\tanh(x/2)$: it is Tran and Van's continuous equation with x replaced by $\tau(x)$ in the term F/x. (1) The quantity $W = \tau(m + F_x) - 2F$ is nonnegative for the transform of any nonnegative sequence of mass m and propagates along characteristics with a nonnegative source; it is the discrete analogue of Van's half-slope invariant [5]. (2) It gives the curvature barrier $\tau F_{xx} > -1$ for all $m \le 1$; Jang and Tran's version of this barrier (their Lemma 3.6) needed $m < 1/2$. (3) Characteristics leave the half-line through $x = 0$; with the barrier they never cross, so the Hamilton–Jacobi equation has a global classical solution, with slope m at $x = 0$ by an ODE argument along characteristics. (4) Finite truncations of (1) are positive and conserve mass exactly; their transforms are supersolutions; transport comparison puts them above the characteristic solution, so a limit point cannot lose mass. (5) With constant mass, the number of clusters is explicit and (1) becomes lower triangular: uniqueness, relaxation and part (i) follow.

1 The closed triangular system and uniqueness

Proposition 7. Let ρ be a mass-conserving solution of mass $m > 0$ and $n_0(t) = \sum_n \rho_n(t)$. Then $n_0(t) = m(1-m) + e^{-t/2}(n_0(0) - m(1-m))$; every ρ_n is C^∞ on $[0,\infty)$ and, with $c_n = mn + \frac{n+1}{2}$,

$$\dot\rho_n + c_n\rho_n = n_0(t) - \sum_{j<n}\rho_j + \frac{1}{2}\sum_{i=1}^{n-1} i(n-i)\rho_i\rho_{n-i} \qquad (t \ge 0); \tag{8}$$

and two mass-conserving solutions with the same datum coincide.

Proof. All series converge absolutely because $\sum n\rho_n = m$. With $g = \mathbf{1}_{\{n\}}$ in (3), ρ_n is absolutely continuous and $\dot\rho_n = K(\rho)_n$ almost everywhere, with coagulation loss $nm\rho_n$. With $g \equiv 1$ the right side of (3) is $-\frac{m^2}{2} + \frac{1}{2}\sum(n-1)\rho_n = \frac{1}{2}(m - m^2 - n_0)$; this linear equation for the absolutely continuous n_0 gives the formula (Jang–Tran (1.6)). Since $\sum_{j>n}\rho_j = n_0 - \sum_{j\le n}\rho_j$, (8) holds almost everywhere. Its right side involves only the smooth n_0 and $\rho_1, \dots, \rho_{n-1}$. By induction on n, ρ_n equals its variation-of-constants formula with a continuous, then C^∞, forcing; so $\rho_n \in C^\infty$ and (8) holds everywhere. Two solutions with the same datum have the same n_0, and (8) for $n = 1, 2, \dots$ are linear scalar equations with identical forcings, by induction. □

Jang and Tran proved this uniqueness (their Proposition 1.2) through uniqueness of

viscosity solutions; the triangular structure gives it directly.

2 The auxiliary Hamilton–Jacobi equation

Throughout this section $m \in (0,1]$ and $\rho^0 \geq 0$ with $\sum n\rho_n^0 = m$ are fixed. Put

$$\tau(x) = 2\tanh\frac{x}{2}, \qquad k = \tau' = 1 - \frac{\tau^2}{4} \in (0,1],$$

$$H_m(x,u,p) = \frac{1}{2}(p-m)(p-m-1) + \frac{u}{\tau(x)} - m, \qquad F_0(x) = \sum_n (1-e^{-nx})\rho_n^0.$$

For $x > 0$: $0 < \tau(x) \leq x$, $\tau < 2$, and $1/\tau(x) = \frac{1}{2} + \frac{1}{e^x - 1}$. The function F_0 is C^∞ on $(0,\infty)$, positive, with $0 \leq F_0' = \sum ne^{-nx}\rho_n^0 \leq m$, $F_0'' \leq 0$ and $F_0(x)/x \uparrow m$ as $x \downarrow 0$ (monotone convergence). We study

$$F_t + H_m(x, F, F_x) = 0 \quad (x > 0,\ t \geq 0), \qquad F(\cdot, 0) = F_0. \tag{9}$$

We do not assume that its solution is the transform of a nonnegative sequence: it only serves as a lower barrier in Section 3.

Lemma 10. *$W_0 := \tau(m + F_0') - 2F_0 \geq 0$ and $-m/e \leq \tau F_0'' \leq 0$ on $(0,\infty)$.*

Proof. With $z = e^{-x}$, $\tau = 2(1-z)/(1+z)$. For an integer $j \geq 1$, $(1+z)(1+z+\cdots+z^{j-1}) = 1 + 2(z + \cdots + z^{j-1}) + z^j$, hence

$$\tau j(1+z^j) - 2(1-z^j) = \frac{2(1-z)}{1+z}\left(j(1+z^j) - (1+z)\,\frac{1-z^j}{1-z}\right)$$

$$= \frac{2(1-z)}{1+z}\sum_{i=1}^{j-1}(1-z^i)(1-z^{j-i}) \geq 0.$$

Multiply by ρ_j^0 and sum, using $\sum j\rho_j^0 = m$. Next, $-\tau F_0'' = \sum_j j\rho_j^0\,(\tau j e^{-jx})$ and $\tau j e^{-jx} \leq x j e^{-jx} \leq 1/e$. □

Characteristics. For a label $y > 0$ let $(X,P,U)(t) = (X,P,U)(t;y)$ solve

$$\dot X = P - m - \tfrac{1}{2}, \qquad \dot P = \frac{k(X)}{\tau(X)^2}\,U - \frac{P}{\tau(X)}, \qquad \dot U = \tfrac{1}{2}\,(P^2 + m - m^2) - \frac{U}{\tau(X)}, \tag{11}$$

with $(X,P,U)(0) = (y, F_0'(y), F_0(y))$. These are $\dot X = \partial_p H_m$, $\dot P = -\partial_x H_m - p\,\partial_u H_m$, $\dot U = p\,\partial_p H_m - H_m$. The vector field is C^∞ on $\{X > 0\} \times \mathbb{R}^2$; $[0, t_e(y))$ is the maximal interval of existence. Put $q = m - P$, $v = q + \frac{1}{2}$, $W = \tau(X)(m+P) - 2U$, $D = U/\tau(X) - P$, $\lambda = U/\tau(X)$, $\eta = m - \lambda$, $\varepsilon = \tau(X)^2 v/4$. Direct differentiation gives the exact identities

$$\dot W + \frac{2+k}{2\tau}\,W = q(2m-q)(1-k), \qquad 1 - k = \frac{\tau^2}{4}, \tag{12}$$

$$W = \tau(q - 2D), \tag{13}$$

$$\dot\lambda = \frac{(P-\lambda)^2 + (m-\lambda)(1-m+\lambda)}{2\tau} - \frac{\varepsilon\lambda}{\tau}, \qquad \dot\eta = -\frac{(\eta - q)^2 + \eta(1-\eta)}{2\tau} + \frac{\varepsilon\lambda}{\tau}. \tag{14}$$

In the continuous case ($\tau = x$, $k = 1$) the source in (12) vanishes and (12) is Van's equation (3.3); here it is nonnegative.

Lemma 15 (invariant region)**.** For every $y > 0$ and $t \in [0, t_e(y))$: $0 \le P \le m$, $U > 0$, $W \ge 0$. Consequently $0 < U \le m\tau(X) < 2m$, $0 \le \lambda \le \frac{m+P}{2} \le m$, $-(m+\frac{1}{2}) \le \dot X \le -\frac{1}{2}$, $y/(m+\frac{1}{2}) \le t_e(y) \le 2y$, and $X(t) \downarrow 0$ as $t \uparrow t_e(y)$.

Proof. Fix $T < t_e(y)$; on $[0,T]$, $X \ge \delta > 0$ and $|P|, |U| \le A$. Let $\delta' = \tau(\delta)$ and $V = P^- + (P-m)^+ + U^- + W^-$, where $a^\pm = \max(\pm a, 0)$. V is Lipschitz and almost everywhere $V' = -\dot P \mathbf{1}_{P<0} + \dot P \mathbf{1}_{P>m} - \dot U \mathbf{1}_{U<0} - \dot W \mathbf{1}_{W<0}$. On $\{P<0\}$, $-\dot P = -kU/\tau^2 + P/\tau \le U^-/\delta'^2$. On $\{P>m\}$, since $2U = \tau(m+P) - W \le \tau(m+P) + W^-$ and $k \le 1$, $\dot P \le \frac{k(m+P)-2P}{2\tau} + \frac{W^-}{2\tau^2} \le \frac{m-P}{2\tau} + \frac{W^-}{2\delta'^2} \le \frac{W^-}{2\delta'^2}$. On $\{U<0\}$, $-\dot U = -\frac{1}{2}(P^2 + m(1-m)) + U/\tau < 0$; this uses $m \le 1$. On $\{W<0\}$, (12) gives $-\dot W \le -q(2m-q)(1-k)$, and $q(2m-q) \ge 0$ if $0 \le P \le m$, $q(2m-q) = m^2 - (P^-)^2 \ge -AP^-$ if $P < 0$, $q(2m-q) = -(P-m)^+(2m + (P-m)^+) \ge -(2m+A)(P-m)^+$ if $P > m$. Hence $V' \le LV$ almost everywhere for a constant L, and $V(0) = 0$ by Lemma 10. Grönwall gives $V \equiv 0$. Then $U \le \tau(m+P)/2 \le m\tau$ and $\lambda \le (m+P)/2$. From $\dot U \ge -U/\tau$, $U(t) \ge F_0(y) \exp\!\left(-\int_0^t ds/\tau(X)\right) > 0$. The bounds on $\dot X$ give the bounds on t_e. If X stayed above some $\delta > 0$, the solution would remain in a compact subset of the domain and could be continued; as X decreases, $X \to 0$. □

Lemma 16 (strip condition and curvature barrier)**.** Let $J = \partial_y X$, $\Pi = \partial_y P$, $\Upsilon = \partial_y U$ on $\Omega = \{(y,t) : y > 0,\ 0 \le t < t_e(y)\}$. Then $\Upsilon = PJ$ and $J > 0$ on Ω, and $\Gamma := \tau(X)\Pi/J$ satisfies $-1 < \Gamma \le \sqrt{2m}$.

Proof. Differentiating (11) in y, $A = \Upsilon - PJ$ satisfies $\dot A = -\partial_u H_m\, A = -A/\tau$, and $A(0) = F_0'(y) - F_0'(y) = 0$; so $A \equiv 0$. With $\kappa = 1/\tau$ this gives $\dot J = \Pi$ and $\dot\Pi = -J(\kappa'' U + 2\kappa' P) - \kappa\Pi$. Let $t_1 = \sup\{t < t_e : J > 0 \text{ on } [0,t]\}$. On $[0,t_1)$, $r = \Pi/J$ satisfies $\dot r = -r^2 - \kappa r - 2\kappa' P - \kappa'' U$, and $\Gamma = \tau r$ satisfies

$$\tau\dot\Gamma = -\Gamma^2 - \Gamma(1 + k(q+\tfrac{1}{2})) - 2kD. \tag{17}$$

By (13) and Lemma 15, $D \le q/2$; also $D \ge -P \ge -m$. Initially $\Gamma(0) = \tau(y)F_0''(y) \in [-m/e, 0]$. At a first time with $\Gamma = -1$, (17) gives $\tau\dot\Gamma = k(q + \frac{1}{2} - 2D) \ge k/2 > 0$, which is impossible; at a first time with $\Gamma = \sqrt{2m}$ it gives $\tau\dot\Gamma \le -2m - \sqrt{2m} + 2km < 0$, also impossible. So $-1 < \Gamma \le \sqrt{2m}$ on $[0,t_1)$. If $t_1 < t_e$, then $X \ge X(t_1) > 0$ on $[0,t_1]$, so $r \ge -1/\tau(X(t_1))$ and $J(s) = \exp\int_0^s r \ge \exp(-t_1/\tau(X(t_1)))$ on $[0,t_1)$; by continuity $J > 0$ on a neighbourhood of t_1, a contradiction. Hence $J > 0$ on Ω. □

With only the bound $D \le q$ (which is what $F \le m(1-e^{-x})$ gives) the right side of (17) at $\Gamma = -1$ is only known to be at least $k(\frac{1}{2} - q)$, positive when $q < \frac{1}{2}$: this is the origin of the restriction $m < 1/2$ in Jang and Tran's Lemma 3.6. The half-slope invariant is used exactly here.

Proposition 18 (global classical solution)**.** For each $t \ge 0$, $\Omega_t = \{y : t_e(y) > t\}$ is an interval $(a(t), \infty)$, with $a(0) = 0$ and $a(t) \ge t/2$, and $X(t;\cdot)$ maps it increasingly onto $(0,\infty)$. The map $\Phi(y,t) = (X(t;y), t)$ is a C^∞ diffeomorphism of Ω onto $(0,\infty) \times [0,\infty)$. The function $F = U \circ \Phi^{-1}$ is C^∞ on $(0,\infty) \times [0,\infty)$ (one-sided derivatives at $t = 0$), $F(\cdot,0) = F_0$, $F_x = P \circ \Phi^{-1}$, F solves (9), and

$$0 \le F_x \le m, \qquad 0 < F \le m\tau(x), \qquad \tau(m+F_x) - 2F \ge 0, \qquad -1 < \tau F_{xx} \le \sqrt{2m}.$$

Proof. Ω is relatively open (lower semicontinuity of t_e) and Φ is C^∞ with Jacobian determinant $J > 0$. *Interval property.* Let $y < y'$, $y \in \Omega_t$, and $s^* = \sup\{s \le t : [y, y'] \times [0, s] \subset \Omega\}$. For $s < s^*$ and $\eta \in [y, y']$, $X(s;\eta) \ge X(s;y) \ge X(t;y) > 0$, because $J > 0$ and X decreases in time. So on $[0, s^*)$ these characteristics stay in the compact set $[X(t;y), y'] \times [0, m] \times [0, 2m]$, near which the vector field is smooth; a uniform extension time shows $s^* = t$ and $y' \in \Omega_t$. Thus $\Omega_t = (a(t), \infty)$. As $t_e(y) \le 2y$, $a(t) \ge t/2$. *Onto.* $X(t;y) \ge y - (m + \frac{1}{2})t \to \infty$. If $X(t;y) \ge \ell > 0$ on Ω_t, then $a = a(t) > 0$, $t_e(a) \le t$, and continuous dependence gives, for $s < t_e(a)$, $X(s;a) = \lim_{y \downarrow a} X(s;y) \ge \ell$, contradicting $X(s;a) \to 0$ (Lemma 15). So Φ is a bijection with positive Jacobian, hence a diffeomorphism. *Equation.* Differentiating $U(t;y) = F(X(t;y), t)$ in y gives $\Upsilon = F_x J$, so $F_x = P$ by Lemma 16; in t, $\dot U = F_x \dot X + F_t$, so $F_t = \dot U - P\partial_p H_m = -H_m$. Differentiating $F_x(X,t) = P$ in y gives $F_{xx} = \Pi/J$. The bounds are Lemmas 15 and 16. □

Lemma 19 (comparison). Let $G \in C^1((0,\infty)\times[0,\infty))$ satisfy $G_t + H_m(x, G, G_x) \ge 0$, $0 \le G_x \le m$ and $G(\cdot, 0) \ge F_0$. Then $G \ge F$.

Proof. Since $H_m(x,u,p) - H_m(x,\tilde u,\tilde p) = (p-\tilde p)(\frac{p+\tilde p}{2} - m - \frac{1}{2}) + (u-\tilde u)/\tau$, $w = F - G$ satisfies $w_t + b w_x + w/\tau \le 0$ with $b = \frac{F_x + G_x}{2} - m - \frac{1}{2} \in [-(m+\frac{1}{2}), -\frac{1}{2}]$ continuous. Fix $x_1, t_1 > 0$ and let Y solve $\dot Y = b(Y,t)$, $Y(t_1) = x_1$, backwards in time (Peano). Backwards, Y increases: $Y(t) \in [x_1 + \frac{t_1 - t}{2}, x_1 + (m+\frac{1}{2})(t_1 - t)]$, so Y exists on $[0, t_1]$ inside $(0,\infty)$. Then $g(t) = w(Y(t),t)$ satisfies $\dot g \le -g/\tau(Y)$, so $w(x_1,t_1) \le w(Y(0),0)\exp\Big(-\int_0^{t_1} 1/\tau(Y)\Big) \le 0$. No boundary condition at $x = 0$ is needed. □

Lemma 20 (slope at the boundary). (a) For $m \in (0,1]$ and every $t \ge 0$, $F(x,t)/x \to m$ as $x \downarrow 0$. (b) If $m < 1$, $t_0 > 0$ and $0 < \delta < \min\{1, t_0/2\}$, then for $0 < x < \delta$ and $t \ge t_0$

$$0 \le m - \frac{F(x,t)}{\tau(x)} \le K x^\alpha,$$

$$\alpha = \frac{1-m}{2m+1}, \qquad K = m\delta^{-\alpha} + \frac{2C}{2-\alpha}\,\delta^{2-\alpha}, \qquad C = \frac{m(m+\frac{1}{2})}{4}.$$

Proof. At $t = 0$, (a) is $F_0(x)/x \to m$. Let $t > 0$, $0 < \delta < \min\{1, t/2\}$, $0 < x < \delta$, and follow the characteristic through (x,t). Its label is $y \ge x + t/2 > \delta$ and X decreases at speed at least $\frac{1}{2}$, so $X(s_1) = \delta$ for some $s_1 \in [t - 2\delta, t) \subset (0,t)$, and $X \in [x, \delta]$ on $[s_1, t]$. Regard λ, η on this segment as functions of X; $d/dX = -v^{-1} d/ds$, $\frac{1}{2} \le v \le m + \frac{1}{2} \le \frac{3}{2}$, $\varepsilon \le \frac{3}{8} X^2$, and $\varepsilon\lambda/\tau \le CX$.

(b) As $\eta \le m < 1$, $\eta(1-\eta) \ge (1-m)\eta$, and (14) gives $\dot\eta \le -\frac{(1-m)\eta}{2\tau} + CX$. Since $\eta \ge 0$ and $\tau \le X$, $\frac{d\eta}{dX} \ge \frac{(1-m)\eta}{2v\tau} - \frac{CX}{v} \ge \frac{\alpha}{X}\,\eta - 2CX$. Integrating $(X^{-\alpha}\eta)' \ge -2CX^{1-\alpha}$ over $[x, \delta]$ with $\eta(\delta) \le m$ gives the bound, whose constant does not depend on $t \ge t_0$.

(a) Let $m \le 1$ and $\theta \in (0, m)$; choose δ also with $\frac{3}{8}\delta^2 \le \frac{\theta}{4}$. As $1 - m + \lambda \ge \lambda$, (14) gives $\dot\lambda \ge \frac{\lambda}{\tau}(\frac{m-\lambda}{2} - \varepsilon)$. So on $[s_1, t]$, $\dot\lambda \ge \theta\lambda/(4\tau) > 0$ whenever $\lambda \le m - \theta$: the set $\{\lambda \ge m - \theta\}$ is never left once entered, and while $\lambda \le m - \theta$, $d\lambda/dX \le -\frac{\theta}{6}\,\lambda/X$,

i.e. $\lambda(X) \geq \lambda(\delta)(\delta/X)^{\theta/6}$. The entry value satisfies $\lambda(\delta) = F(\delta, s_1)/\tau(\delta) \geq \lambda_* := \min_{s\in[0,t]} F(\delta, s)/\tau(\delta) > 0$. Hence $F(x,t)/\tau(x) \geq m - \theta$ for $x < \delta(\lambda_*/(m-\theta))^{6/\theta}$, while $F/\tau \leq m$. Since θ is arbitrary and $\tau(x)/x \to 1$, (a) follows. □

3 Existence: positive truncations and comparison

For $L \geq 2$ the truncated system keeps the coagulation events with $i + j \leq L$:

$$\dot\rho_n^L = \frac{1}{2}\sum_{i=1}^{n-1} i(n-i)\rho_i^L\rho_{n-i}^L - n\rho_n^L\sum_{j=1}^{L-n} j\rho_j^L - \frac{n-1}{2}\rho_n^L + \sum_{j=n+1}^{L}\rho_j^L, \qquad 1 \leq n \leq L, \quad (21)$$

with $\rho_n^L(0) = \rho_n^0$ for $n < L$ and $\rho_L^L(0) = \rho_L^0 + L^{-1}\sum_{j>L} j\rho_j^0$. Let $F^L(x,t) = \sum_{n\leq L}(1 - e^{-nx})\rho_n^L(t)$.

Lemma 22. (21) has a unique global solution; it is nonnegative and $\sum_{n\leq L} n\rho_n^L \equiv m$. Moreover F^L is C^1, $0 \leq F_x^L \leq m$, $F^L(\cdot, 0) \geq F_0$, and

$$F_t^L + H_m(x, F^L, F_x^L) = E_L := \frac{1}{2}\sum_{\substack{i,j\leq L\\ i+j>L}} ij\rho_i^L\rho_j^L(1 - e^{-ix})(1 - e^{-jx}) \geq 0. \quad (23)$$

Proof. The right side of (21) is polynomial and quasi-positive ($\rho^L \geq 0$, $\rho_n^L = 0$ imply $\dot\rho_n^L \geq 0$), so the nonnegative orthant is invariant. Coagulation conserves $i + j$ and fragmentation conserves size, so $\sum n\rho_n^L$ is constant, equal to $\sum_{n<L} n\rho_n^0 + L\rho_L^0 + \sum_{j>L} j\rho_j^0 = m$; the solution is bounded, hence global. For (23) take $g(n) = 1 - z^n$, $z = e^{-x}$, in the weak form of (21). As $g(i+j) - g(i) - g(j) = -(1-z^i)(1-z^j)$, coagulation contributes $-\frac{1}{2}(\sum_n n(1-z^n)\rho_n^L)^2 + E_L = -\frac{1}{2}(m - F_x^L)^2 + E_L$. For fragmentation, $\sum_{k=1}^{n-1}(g(n) - g(k) - g(n-k)) = -(n-1)(1+z^n) + 2(z - z^n)/(1-z)$; summing against $-\frac{1}{2}\rho_n^L$ with $\sum z^n\rho_n^L = n_0 - F^L$, $\sum nz^n\rho_n^L = F_x^L$ gives $\frac{1}{2}(m + F_x^L) - F^L(\frac{1}{1-z} - \frac{1}{2}) = \frac{1}{2}(m + F_x^L) - F^L/\tau$. Hence $F_t^L = -\frac{1}{2}(m - F_x^L)^2 + \frac{1}{2}(m + F_x^L) - F^L/\tau + E_L$, which is (23). The same computation with $E = 0$ gives Jang–Tran's (1.7) for a mass-conserving solution. Finally $F^L(x,0) - F_0(x) = \sum_{j>L} j\rho_j^0(\frac{1-e^{-Lx}}{L} - \frac{1-e^{-jx}}{j}) \geq 0$, since $s \mapsto (1 - e^{-sx})/s$ decreases. □

Proof (of Theorem 4). Uniqueness and the formula for n_0 are Proposition 7.

Barrier. By Lemmas 19 and 22, $F^L \geq F$ for every L.

Compactness. Extend $\rho_n^L = 0$ for $n > L$. Then $0 \leq \rho_n^L \leq m/n$ and $|\dot\rho_n^L| \leq 3m^2/2 + 3m/2$. By Arzelà–Ascoli and a diagonal argument, along some $L_k \to \infty$, $\rho_n^{L_k} \to \rho_n$ locally uniformly on $[0,\infty)$ for every n, with $\rho_n \geq 0$ continuous, $\rho_n(0) = \rho_n^0$, and $M(t) := \sum n\rho_n(t) \leq m$ by Fatou. As $\sum_{n>N}\rho_n^L \leq m/(N+1)$, $F^{L_k}(x,t) \to F^\rho(x,t) := \sum(1 - e^{-nx})\rho_n(t)$.

No mass loss. $F^\rho \geq F$, and $1 - e^{-nx} \leq nx$ gives $M(t) \geq F^\rho(x,t)/x \geq F(x,t)/x$ for all $x > 0$. By Lemma 20(a), $M(t) \geq m$; so $M(t) = m$ for every $t \geq 0$.

Limit equation. Fix t. Coordinatewise convergence and equality of the masses give tightness: if $\sum_{n\leq N} n\rho_n(t) > m - \epsilon$, then $\sum_{n>N} n\rho_n^{L_k}(t) < 2\epsilon$ for large k. So $\mu_n^{L_k}(t) = \sum_{j\leq L_k - n} j\rho_j^{L_k}(t) \to m$ for fixed n. In the integral form of (21), the coagulation gain is a finite sum, the loss $n\rho_n^L\mu_n^L \leq m^2$ converges pointwise to $nm\rho_n$ (dominated convergence),

and the fragmentation gain converges uniformly by the number-tail bound. Hence $\rho_n(t) = \rho_n^0 + \int_0^t K(\rho(s))_n\,ds$ for all n and t.

Regularity. If $t_j \to t$, the nonnegative sequences $\rho(t_j)$ converge coordinatewise to $\rho(t)$ with equal masses, so $\sum n|\rho_n(t_j) - \rho_n(t)| = 2\sum n(\rho_n(t) - \rho_n(t_j))^+ \to 0$ (discrete Scheffé, dominated convergence): $\rho \in C([0,\infty);\ell^1_1)$. The map $K : \ell^1_1 \to \ell^1$ is locally Lipschitz (bilinear coagulation part of norm at most $\frac{3}{2}\|u\|_{\ell^1_1}\|v\|_{\ell^1_1}$, linear fragmentation part of norm at most $\frac{3}{2}$). So the integral identity holds in ℓ^1 with continuous integrand, $\rho \in C^1([0,\infty);\ell^1)$, and for bounded g, $\sum g(n)\rho_n$ is C^1 with derivative $\sum g(n)K(\rho)_n$, the right side of (3) after rearranging absolutely convergent double series. Thus ρ is a classical mass-conserving solution; Proposition 7 gives C^∞ components. By uniqueness the whole family ρ^L converges. □

Remark 24. The transform of the solution is the characteristic solution: $F^\rho = F$. Indeed F^ρ is C^1 and solves (9) by the computation of Lemma 22 with $E = 0$; $w = F^\rho - F$ then solves $w_t + bw_x + w/\tau = 0$, $w(\cdot,0) = 0$, and the argument of Lemma 19 applied to $\pm w$ gives $w = 0$. This is not needed above. ⌟

4 Stationary states and relaxation

Proof (of Theorem 5)**.** Let $0 < m \le 1$ and ρ be the solution of Theorem 4 for a datum of mass m. In (8), $n_0(t) \to m(1-m)$. If $\dot u + cu = f(t)$ with $c > 0$ and $f(t) \to f_\infty$, then $u(t) \to f_\infty/c$. Applied to (8) for $n = 1, 2, \dots$ in turn, this gives $\rho_n(t) \to \tilde\rho_n$, where $c_n\tilde\rho_n = m(1-m) - \sum_{j<n}\tilde\rho_j + \frac{1}{2}\sum i(n-i)\tilde\rho_i\tilde\rho_{n-i}$, which is (2) since $2c_n = (2m+1)n+1$. Hence $\tilde\rho_n \ge 0$. For $m = 1$ the recursion gives $\tilde\rho \equiv 0$.

Let now $m < 1$. By the number-tail bound, $F^\rho(x,t) \to \tilde F(x) := \sum(1 - e^{-nx})\tilde\rho_n$ and $\sum\tilde\rho_n = \lim n_0(t) = m(1-m)$. By the barrier step of the previous proof, $F^\rho \ge F$, and Lemma 20(b) with $t_0 = 1$ gives $F^\rho(x,t) \ge \tau(x)(m - Kx^\alpha)$ for $0 < x < \delta$, $t \ge 1$. Letting $t \to \infty$, $\tilde F(x) \ge \tau(x)(m - Kx^\alpha)$, so $\sum n\tilde\rho_n = \lim_{x\downarrow 0}\tilde F(x)/x \ge m$; Fatou gives the reverse inequality. With these two sums, $\sum_{j>n}\tilde\rho_j = m(1-m) - \sum_{j\le n}\tilde\rho_j$, and (2) is the equation $K(\tilde\rho)_n = 0$: $\tilde\rho$ is stationary. Strict positivity: $\tilde\rho_1 = m(1-m)/(m+1) > 0$, and the numerator of (2) equals $2\sum_{j\ge n}\tilde\rho_j + \sum_{i=1}^{n-1} i(n-i)\tilde\rho_i\tilde\rho_{n-i}$, which is at least $(n-1)\tilde\rho_1\tilde\rho_{n-1} > 0$ if $\tilde\rho_{n-1} > 0$. Uniqueness: a stationary σ with $m_1(\sigma) = m$ is a constant mass-conserving solution; Proposition 7 gives $\sum\sigma_n = m(1-m)$ and (8) with $\dot\sigma = 0$, that is (2); so $\sigma = \tilde\rho$. Relaxation: $\rho_n(t) \to \tilde\rho_n$ and both masses equal m, so $\rho(t) \to \tilde\rho$ in ℓ^1_1 by the discrete Scheffé argument. □

Jang and Tran obtained the stationary state for $m \le 1/2$ as the long-time limit of their solutions (Lemma 5.7, using their Lemma 5.1 on viscosity solutions); for $m \ge 1$ none exists (their Lemma 5.6).

5 Decay of the stationary state

Attribution. The singularity analysis of this section was found by GPT-6-Astra; it originally served a continuation-in-m proof of positivity. Here positivity and the moment identities are supplied by Theorem 5, so only the fixed-m analysis is needed; it was rechecked and condensed by the final author.

Fix $m \in (0,1)$, $a = 2m+1$, $N = m(1-m)$, $H(z) = \sum\tilde\rho_n z^n$, $P(z) = zH'(z) = \sum n\tilde\rho_n z^n$, with common radius R. By Theorem 5, $R \ge 1$, P is strictly increasing on $[0,1]$ with $P(1) = m$, $H(1) = N$. Multiplying $K(\tilde\rho)_n = 0$ by $2z^n$ and summing (absolutely

convergent) gives, for $|z| < R$,

$$(1-z)(aP - P^2) - 2zN + (1+z)H = 0, \tag{25}$$

and, after differentiation and elimination of H,

$$z(1-z^2)(a-2P)P' = 2z(m-P)(P+1-m) - (1-z)^2 P. \tag{26}$$

Lemma 27. *H and P extend analytically across $z = 1$; hence $R > 1$.*

Proof. For $0 < z < 1$ put $r = (m-P)/(1-z) > 0$. Substituting into (26) gives

$$r' = \frac{m - r + 2z(z+2)r^2}{z(1+z)(1+2(1-z)r)}.$$

Suppose r is unbounded near 1. For z close to 1 the numerator is positive once r exceeds a fixed constant, so r eventually increases to $+\infty$; since $(1-z)r = m - P \to 0$, the denominator tends to 2 and $r'/r^2 \to 3$. Integrating $(1/r)' = -r'/r^2$ up to 1 gives $(1-z)r \to 1/3$, a contradiction. So r is bounded, r' is bounded, r has a limit r_1, and the equation is analytic at $(1, r_1)$ (denominator 2); its analytic solution through that point coincides with r. Thus $P = m - (1-z)r$ and, by (25), $H = N - \frac{1-z}{1+z}(N + aP - P^2)$ are analytic at 1. Pringsheim's theorem Theorem IV.6 [6] (the coefficients are positive) excludes $R = 1$. □

Lemma 28. *$R < \infty$, $P(R^-) = p_* := m + \frac{1}{2}$, and near R, $P(z) = p_* - b\sqrt{1 - z/R} + O(1 - z/R)$ with a convergent expansion in powers of $\sqrt{1 - z/R}$ and b as in Theorem 6. The point R is the only singularity of H on $|z| = R$.*

Proof. On $(1, R)$, $P > P(1) = m$. If P reached p_* at some $z_0 < R$, the left side of (26) would vanish there while the right side equals $-\frac{3}{2}z_0 - (1-z_0)^2 p_* < 0$; so $m < P < p_*$ on $(1, R)$. For $z \geq 2$ in that interval, both sides of (26) are negative and $P' \geq \frac{(z-1)P}{z(z+1)(a-2P)} \geq \frac{P}{3z}$ (as $0 < a - 2P < 1$); if $R = \infty$ this contradicts $P < p_*$. If $P(R^-) < p_*$, (26) is regular at $(R, P(R^-))$ and continues P analytically across R, contradicting Pringsheim; so $P(R^-) = p_*$. Near this point the inverse function solves the regular equation $dz/dp = B(z)(a-2p)/D(z,p)$, $B = z(1-z^2)$, $D = 2z(m-p)(p+1-m) - (1-z)^2 p$, with $B(R) < 0$, $D(R, p_*) < 0$; so $z(p)$ is analytic at p_* with $z'(p_*) = 0$, $z''(p_*) = -2B(R)/D(R, p_*) < 0$. Inverting this quadratic critical point gives the expansion, with $b^2 = RD(R, p_*)/B(R)$, which is the stated formula. Finally $\sum n\tilde{\rho}_n R^n = p_* < \infty$, so H, P are continuous on $|z| \leq R$. If $|w| = R$, $w \neq R$, strict positivity of $\tilde{\rho}_1, \tilde{\rho}_2$ gives $|P(w)| < P(R) = a/2$, so $a - 2P(w) \neq 0$ and $w \neq 1$; then (25) defines P analytically in terms of (z, H) near $(w, H(w), P(w))$, and the regular equation $H' = P/z$ continues H analytically across w. □

Proof (of Theorem 6). By Lemma 28 and compactness, H is analytic in a Δ-domain at R Definition VI.1, Section VI.3 [6]. With $s = 1 - z/R$, $dH/ds = -P/(1-s)$, so integrating the expansion of P gives $H = A(s) + s^{3/2}g(s)$ with A, g analytic near 0 and $g(0) = \frac{2b}{3}$. Hence $H - A_0 - A_1 s - A_2 s^2 - \frac{2b}{3}s^{3/2} = O(s^{5/2})$ in the Δ-domain, where $A_0 + A_1 s + A_2 s^2$ is a polynomial in z and does not contribute to coefficients of index $n > 2$. The transfer theorems Theorems VI.1 and VI.3 [6] give $[z^n]H = \frac{2b}{3\Gamma(-3/2)} R^{-n} n^{-5/2}(1 + O(n^{-1}))$, and $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$. □

In particular $\tilde{\rho}$ has exponential moments $\sum q^n \tilde{\rho}_n < \infty$ for $q < R(m)$. The slow decay for m close to 1 (where $R(m) \to 1$) is visible numerically: for $m = 0.75$, the mass carried by sizes above 3000 is about 2×10^{-9}.

6 Remarks and limitations

Remark 29 (what is used where)**.** The restriction $m \le 1$ enters only through the face $U = 0$ of the invariant region ($\dot{U} = \frac{1}{2}(P^2 + m - m^2)$ there); for $m > 1$ Jang and Tran's Theorem 1.1 excludes global mass-conserving solutions, and Proposition 7 shows directly that n_0 would become negative. The restriction $m < 1$ enters only in the uniform-in-time estimate Lemma 20(b), which is needed for the stationary state. ⌟

Remark 30 (moments)**.** Theorem 4 does not assert that $m_2(\rho(t))$ is finite for $t > 0$ when $m_2(\rho^0) = \infty$; it is not needed. The solution is classical in Jang and Tran's sense (each component is differentiable and satisfies (1)). ⌟

Remark 31 (relation to the literature)**.** Tran and Van [4], Jang and Tran, and Van construct solutions as vanishing-viscosity limits of viscosity solutions and then prove a Bernstein (positivity) property, which requires regularity of the Hamilton–Jacobi solution at the boundary $x = 0$. Here positivity comes from the finite truncations, the characteristic solution is used only as a lower barrier, and its boundary behaviour reduces to the ODE inequality of Lemma 20. The invariant W is the discrete form of Van's half-slope invariant $2M - xM_x \ge 0$, $M = mx - F$ [5]. ⌟

Remark 32 (an alternative stationary argument)**.** GPT-6-Astra also sketched a purely stationary proof of positivity of (2), by continuation in m from Jang and Tran's range $m \le 1/2$, based on a parameter-uniform version of Lemmas 27 and 28 and uniform singularity transfer. Its uniformity step (a common Δ-domain and uniform transfer near the moving turning point $R(m)$) has not been checked to the standard of this paper, and it is not used; the draft is kept outside the deliverables. ⌟

7 Computer checks and independent verification

The proofs do not depend on numerical computation. Scripts in `scripts/` (each prints PASS/FAIL): `check_evolution_identities.py` verifies exactly (sympy) every identity of Sections 2–3: $1/\tau = \frac{1}{2} + \frac{1}{e^x - 1}$, the transform equations for the full and truncated systems ((23) with E_L), the characteristic equations, (12), (13), (14), the strip condition, the Riccati equation, (17), the identity of Lemma 10 for $j \le 60$ and the elementary inequalities. `check_identities.py` (GPT-6-Astra) verifies the stationary identities (25), (26), the r-equation, the turning-point quantities and b^2; `check_coefficients_arb.py` encloses the first 256 coefficients of (2) at six masses in Arb ball arithmetic (finite consistency check). `numerics_evolution.py` is floating-point evidence only: invariants and barrier along characteristics for monomer and infinite-second-moment data, the comparison $F^L \ge F$, agreement of the triangular and truncated systems, and the long-time limit. Its complete run is reported in `README.md`.

Independent verification. The points singled out for checking (the no-crossing and global-coverage argument, Lemma 16 and Proposition 18; the uniformity in Lemma 20(a) over characteristics arriving near $x = 0$; the identity (23) and the sign of E_L; the tightness step in the coagulation loss; the use of Lemma 20(b) to prevent mass escape as

$t \to \infty$; the singularity analysis) were re-derived by an independent verifier, who found no mathematical gap. The verifier re-ran all scripts and added independent checks (different integrators, infinite-second-moment data at $m = 0.9$ and $m = 1$, Arb enclosures of (2), an independent computation of $R(m)$ and of the constant of Theorem 6) in `verification/`; the report, with the list of corrections, is `verification.md`.

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